

Team members:

Lenhle Dlamini 202302047

Lindokuhle Nkabinde 202304158

Andiswa Mdluli 202304513

Nsika Mkhathshwa 202401362

Gcinwayinkosi Bhembe 202405187

Problem Statement: Hostel Access System

Hostels often face challenges in managing secure and efficient access for residents, staff, and visitors. Traditional methods such as manual registers, physical keys, or basic ID cards are prone to several issues:

Security Risks: Unauthorized entry due to lost keys, stolen ID cards, or lax manual checks.

Inefficiency: Time-consuming verification processes at entry points, especially during peak hours.

Lack of Monitoring: Difficulty in tracking entry/exit logs, making it hard to identify unusual activity or enforce curfews.

Limited Integration: Existing systems often fail to integrate with digital records, making reporting and oversight cumbersome.

Visitor Management: Temporary access for guests or maintenance staff is poorly controlled, leading to potential breaches.

These challenges result in compromised safety, administrative burdens, and dissatisfaction among hostel residents.

The goal is to design and implement a Hostel Access System that ensures:

Secure, reliable, and automated entry/exit for authorized individuals.

Real-time monitoring and logging of movements.

Easy integration with hostel management databases.

Controlled visitor access with temporary permissions.

Enhanced convenience for residents while maintaining strict security standards.